Modelling LCA Modules C and D for Novel Bio-Based and Waste-Derived 3D-Printed Building Panels

TL;DR

- For a novel biopolymer panel (seagrass, bark flour, cellulose, pea protein) with no established EoL market, the most defensible baseline is a **cut-off / "polluter-pays" attributional model** built around the realistic current fate of unsorted construction waste in your region — most likely **municipal incineration with energy recovery (or landfill)** — reported as Module C, with all speculative recycling/reuse benefits confined to a clearly separated, sensitivity-tested **Module D**. Do NOT fold C and D together.
- Because you have already chosen a biogenic carbon method in Module A, **consistency is the binding constraint**: under EN 15804+A2 the −1 removal booked in A1–A3 must be released as a +1 emission in Module C (C3 for incineration, C4 for landfill), netting biogenic carbon to zero over the life cycle; the standard forbids crediting temporary or permanent biogenic storage.
- Use **scenario + sensitivity analysis as the core method, not a single answer** : report at least landfill, incineration-with-energy-recovery and composting as parallel Module C scenarios, and treat the substitute (grid electricity, fertiliser/peat) and any quality-loss correction factor as explicit sensitivity parameters, since results are dominated by these choices.

Key Findings

1. **Module C captures real EoL processing burdens; Module D captures avoided burdens (credits) beyond the system boundary.** Under EN 15804+A2, C1 = deconstruction/demolition, C2 = transport to waste processing, C3 = waste processing for reuse/recovery/recycling, C4 = final disposal. Net recycling/energy/reuse benefits are reported separately in Module D. The modules must not be added together.
2. **There is no methodological consensus on EoL allocation.** The main competing approaches — cut-off (100:0/recycled-content), avoided burden/substitution (0:100), the 50/50 method, system expansion, and the EU's Circular Footprint Formula (CFF) — can reverse the ranking of materials. The choice should be governed by goal and scope (ISO 14044) and applied consistently.
3. **For a novel material with no recycling infrastructure, the cut-off approach is the most epistemically honest baseline** , because it does not award speculative credits for recycling routes or substitution markets that do not yet exist. Avoided-burden/CFF credits should be presented as upside scenarios in Module D.
4. **Biogenic carbon accounting dominates GWP results for these materials** and must be handled identically in A and C. EN 15804+A2 mandates the −1/+1 convention with a net-zero life-cycle balance; if biogenic carbon physically leaves the system (to recycling/reuse or remains stored), a "virtual" biogenic CO_2 emission is added at EoL so uptakes and emissions balance.
5. **Results are most sensitive to two choices**: (a) the substituted product in Module D (grid electricity mix for energy recovery; fertiliser/peat for compost), and (b) the quality/value-correction factor for recovered material. These should be the primary sensitivity parameters.

Details

Standards framework: EN 15804+A2 vs ISO 14040/14044

EN 15804:2012+A2:2019 is the construction-product Product Category Rule under ISO 14025. It was approved by CEN on 21 July 2019, and after the EN 15804+A1 transition window closed in October 2022 programme operators (e.g. IBU) accept only +A2 for new EPDs; under +A2, EPDs must declare modules C1–C4 and D (previously optional). It splits climate change into GWP-fossil, GWP-biogenic, GWP-luluc and GWP-total, plus an additional GWP-GHG indicator in which the characterisation factor for biogenic CO_2 is set to zero. EN 15804+A2 contains an **informative end-of-life/Module D formula derived from the PEF Circular Footprint Formula**.

A key limitation flagged in the literature (Int. J. LCA, "End-of-life modelling of buildings..."): the EN 15804+A2 formula assumes E_recycling = E_recycled and E_v = E*_v, so it "works well for closed-loop recycling but leads to wrong results for open-loop recycling." Module D in EN 15804 also uses **current average technology** for substitution (not future projections), and reused/recycled material is assumed to enter burden-free (cut-off at the system entry) to avoid double counting between A1–A3 and D.

ISO 14040/14044 is the generic LCA framework. It sets a decision hierarchy for multifunctionality: (1) avoid allocation by subdivision or system expansion; (2) allocate by physical relationship; (3) allocate by other relationships (e.g. economic value). For recycling specifically (clause 4.3.4.3), it describes closed-loop allocation (applies where inherent material properties are maintained, so recycling avoids virgin production) and open-loop allocation, and names cut-off, avoided-burden and 50/50 as applicable procedures — but leaves the choice to the practitioner, requiring only explicit statement, justification by goal/scope and internal consistency. The 2020 Amendment 2 added an annex on system expansion/substitution but, as critics (Heijungs et al., Frontiers in Sustainability 2021) note, did not resolve the confusion about when substitution is permissible in attributional LCA.

Conflicts/gaps for bio-based materials: EN 15804 is prescriptive (fixed −1/+1 biogenic rule, prescribed CFF-derived Module D, no storage credit), whereas ISO 14044 is permissive. ISO permits crediting temporary storage in some interpretations; EN 15804 (clause 5.4.3) explicitly forbids it ("temporary storage of carbon within products shall not be included in the calculations of GWP impacts"). Neither framework provides default CFF parameters for novel biodegradable materials, nor handles the timing of slow biomass regrowth vs fast EoL release — a recognised gap (the −1/+1 approach "oversimplifies... treating growth and emission to occur instantly").

EoL Scenario 1 — Landfill

(A) Module C burdens. Anaerobic degradation of biodegradable fractions generates landfill gas (CH_4 + CO_2); methane is the dominant climate burden. Modelled in C4 using time-bounded gas-generation models (typically 100-year), with assumptions on gas collection efficiency and oxidation in cover soils. Leachate emissions to groundwater contribute toxicity/eutrophication. For slowly degrading biopolymers this matters: per a PLA LCA review, "only one percent will be degraded after 100 years," and NatureWorks' Ingeo study (Polymer Degradation & Stability, 2012) concluded that semicrystalline PLA "will not lead to significant generation of methane" under anaerobic landfill conditions — so much carbon is effectively stored.

(B) Benefits/credits. If landfill gas is captured and combusted for energy, avoided grid energy is credited — in Module D (or within C as net, depending on system model). Carbon that does not degrade represents biogenic carbon storage, but EN 15804+A2 does not allow this to be credited as a removal (see D below).

(C) Methodological approaches. Landfill is typically modelled with a cut-off/disposal approach (burdens to the product generating the waste). The avoided-burden of captured-gas energy is system expansion. A documented debate (Anshassi, Sackles & Townsend review; Int. J. LCA derelict-fishing-gear study): whether landfill or incineration is more GHG-favourable hinges on landfill gas collection efficiency (>75%) and gas-to-energy efficiency (>90%) — under high-capture assumptions landfill can outperform incineration on GWP.

(D) Biogenic carbon. Under EN 15804+A2, biogenic C sequestered in A1–A3 must be released in C4 (landfill is the C4 fate). Even though physically much carbon remains stored in the landfill, the standard requires net-zero biogenic accounting; if carbon is not actually emitted as CO_2, a virtual biogenic CO_2 emission is added at EoL to balance the −1 booked in A. Crediting landfill carbon storage is a recognised pathway that "could lead to undesirable linear pathways... giving the biggest credits for landfilling," contradicting circularity — hence the standard's prohibition.

EoL Scenario 2 — Incineration without energy recovery

(A) Module C burdens. Combustion in C3/C4 oxidises all carbon to CO_2; biogenic CO_2 (booked separately) plus any fossil-derived additives/binders. Auxiliary fuel, flue-gas cleaning, bottom/fly ash disposal. NOx, particulates contribute to acidification, eutrophication, particulate-matter and toxicity categories.

(B) Benefits/credits. None — by definition no energy recovery, so no Module D substitution credit. This is the "worst case" thermal scenario and a useful conservative bound.

(C) Methodological approaches. Straight attributional/cut-off: all burdens to the product system; no system expansion.

(D) Biogenic carbon. Biogenic C released as CO_2 in C3, characterised +1, balancing A1–A3 removal to net zero.

EoL Scenario 3 — Incineration with energy recovery (waste-to-energy)

(A) Module C burdens. As Scenario 2: combustion emissions, ash handling, flue-gas treatment, modelled in C3. Astrup et al. (2014, "Life cycle assessment of thermal waste-to-energy technologies: review and recommendations," Waste Manag.) is the standard methodological reference.

(B) Benefits/credits — Module D. Recovered heat and electricity substitute conventional production; the avoided burden is credited in Module D. In the CFF this is the energy term `(1−B)·R3·(E_ER − LHV·X_ER,heat·E_SE,heat − LHV·X_ER,elec·E_SE,elec)`, where R3 = fraction to energy recovery, LHV = lower heating value, X = recovery efficiencies for heat/electricity, E_SE = emissions of the substituted energy.

(C) Methodological approaches. System expansion/substitution (Module D, or CFF energy term). The B factor (default **B = 0**) assigns full incineration burdens and full substitution credits to the product system; Ekvall et al. ("Factor B in the Circular Footprint Formula") note B=0 implies waste treatment is the determining function, "an assumption [that] is often wrong." Results are highly sensitive to the assumed substituted energy: an ACS Environmental Science & Technology WtE study (Jamesville, NY) found electricity climate impact swung from 0.664–0.951 kg CO_2eq/kWh before system expansion to as low as −0.280 kg CO_2eq/kWh after accounting for avoided waste management — a sign change driven entirely by the substitution assumption.

(D) Biogenic carbon. Biogenic CO_2 released in C3, +1, net zero; only biogenic CH_4 (negligible in combustion) and fossil-carbon additives carry net GWP. Note the GWP-GHG indicator zeroes the biogenic CO_2 characterisation entirely.

EoL Scenario 4 — Composting (aerobic)

(A) Module C burdens. Aerobic decomposition (C3) emits CO_2 (biogenic), plus CH_4 and N_2O from imperfect aeration (significant GWP contributors), and NH_3 (acidification/eutrophication). Process energy for shredding/turning. PLA and many biopolymers degrade poorly under standard composting and can lower compost quality (lactic acid lowering pH, inhibiting germination, per Bandini et al. 2020) — disintegration rate is a key parameter.

(B) Benefits/credits — Module D. Finished compost substitutes synthetic fertiliser (N, P, K) and/or peat, and can sequester some carbon in soil. Credits belong in Module D.

(C) Methodological approaches. System expansion/substitution with Mineral Fertiliser Equivalent (MFE) and peat-substitution factors. Critical: real substitution is far below the 100% often assumed. Andersen et al. (2010c), cited in Boldrin et al., found via interviews that "less than 50% of the citizens using compost in their garden were replacing peat or mineral fertilizers with compost" — i.e. real substitution is "more likely around 50%." Detailed Aarhus/Copenhagen surveys quantified user fractions of ~19–22% for peat, 12–24% for fertiliser and 7–15% for manure (totals ~41–58%). Per a bio-based growing-media LCA review, "MFE for N fertilizers typically varies between 0.2 and 0.4 depending on plant availability... while in most studies the MFE for P and K is almost 1."

(D) Biogenic carbon. Biogenic C released as CO_2 in C3, +1, net zero. Any soil carbon retention — Boldrin et al. (2010, Resources, Conservation and Recycling) "assumed that for compost 14% of the initial carbon was left in the soil after 100 years" — is, under EN 15804+A2, NOT creditable as storage; the virtual-emission rule applies.

EoL Scenario 5 — Mechanical recycling

(A) Module C burdens. Collection, sorting, cleaning, drying, shredding, re-extrusion/re-printing (C3). Bio-based and biodegradable polymers are thermosensitive and hygroscopic (PLA), so reprocessing degrades molecular weight and mechanical properties — downcycling is inherent.

(B) Benefits/credits — Module D. Recovered material substitutes virgin material; credit in Module D, corrected for quality loss.

(C) Methodological approaches. Cut-off (no credit; recyclate enters next system burden-free) vs avoided-burden/substitution vs 50/50 vs CFF material term. For bio-based panels, mechanical recycling rarely returns equivalent material, so a value-correction is essential (see cross-cutting §3).

(D) Biogenic carbon. Carbon transferred in the recyclate physically continues to be stored, but EN 15804+A2 requires a virtual biogenic CO_2 emission at the point material leaves the system, so the studied product still nets to zero; the downstream system books the uptake.

EoL Scenario 6 — Closed-loop recycling (manufacturer take-back, repair, resale)

(A) Module C burdens. Reverse logistics (transport), inspection, repair printing (additive repair of these specific biopolymer panels is demonstrated in CITA's "continual construction" work), refurbishment energy.

(B) Benefits/credits — Module D (or avoided A-stage). Avoided virgin production and avoided new manufacture. For on-site element reuse, RICS guidance allows offsetting A1–A3 and A4; for off-site reuse/recycling, Module D scope is limited to A1–A3. Reuse generally outperforms recycling because, per Sandin & Peters (2018, J. Cleaner Production 184:353–365), "benefits mainly arise due to the avoided production of new products" and so "benefits may not occur in cases with low replacement rates" — the replacement rate is the key factor.

(C) Methodological approaches. Closed-loop allocation (ISO 14044) where properties are maintained — recycling directly avoids virgin production. Replacement/substitution rate and number of additional use cycles are the governing assumptions.

(D) Biogenic carbon. Carbon stays in the loop; net-zero balance over each life cycle; virtual-emission rule applies at the boundary where material exits the studied system.

EoL Scenario 7 — Open-loop recycling (material used in a different product system)

(A) Module C burdens. Collection, reprocessing into a lower-grade product (e.g. panel → filler, insulation, board).

(B) Benefits/credits — Module D. Credit for substituting the material displaced in the receiving system (which may be wood, board, or aggregate, not virgin biopolymer), corrected for quality.

(C) Methodological approaches. This is the hardest allocation case. Options: cut-off, 50/50, avoided-burden, loss-of-quality, market-based (Ekvall), and CFF. Nicholson et al. (MIT) showed cut-off, 50/50 and substitution cluster together while closed-loop and loss-of-quality form a second trend; rankings can switch. The CFF A-factor (0.2–0.8) is meant to reflect supply/demand for the recyclate; EN 15804+A2's simplified formula mis-handles open-loop (see Standards section).

(D) Biogenic carbon. Virtual emission added when biogenic carbon exits to the other product system; studied system nets to zero.

Cross-cutting §1 — Defensible baseline for a novel biopolymer with no EoL market

Model the **realistic current fate of mixed/unsorted construction & demolition waste in your jurisdiction** as the Module C baseline — in most of Europe this is **incineration with energy recovery; in landfill-dominated regions, landfill.** Criteria for scenario selection: (i) current actual waste-management practice, not aspirational routes ("not wishful thinking"); (ii) absence of dedicated collection/sorting for a novel material argues against recycling/composting baselines; (iii) small research quantities → no industrial compost/recycling stream exists; (iv) conservatism — avoid awarding credits for non-existent infrastructure. Present recycling, closed-loop and composting as **prospective Module D scenarios** clearly labelled as contingent on future infrastructure. This aligns with the CITA "Material Stories" LCA approach (Galluccio, Tamke, Nicholas, Svilans, Gaudillière-Jami & Ramsgaard Thomsen, 2025, doi:10.1007/978-3-031-69626-8_3), which used ex-post "cradle-to-construction" analysis with sensitivity analysis to manage exactly this data scarcity for the Radicant panels (bone-glue/glycerol biopolymer with wood flour, bark, seagrass and cotton-bark fillers).

Cross-cutting §2 — Choosing the substitute and its sensitivity

For energy recovery, the substituted electricity/heat is the dominant swing factor. Use **a transparent, regionalised current average grid mix** for an attributional/EN 15804 study (EN 15804 D uses current average technology); a marginal/affected-technology mix is appropriate only for a consequential study, and should be run as a sensitivity. Mathiesen et al. showed that using a single marginal technology is the main error source and recommended using "fundamentally different affected technologies... identified in several possible and fundamentally different future scenarios." For compost, substitute realistic fractions of fertiliser and peat (around 50%, not 100%), using MFE 0.2–0.4 for N. Always run the substitute as a sensitivity parameter; rankings can change with worst/best-case substitution assumptions.

Cross-cutting §3 — Quality loss / value-correction for recycled bio-based materials

Recovered bio-based material is almost always downcycled. Handle via a **value-correction factor** that scales the substitution credit below 1:1. Options: (i) the CFF quality ratios Qs_out/Qp and Qs_in/Qp; (ii) Value-Corrected Substitution (VCS, EMPA 2002 / European Aluminium Association), where the avoided-burden inventory is multiplied by a correction factor β = ratio of secondary to primary market price; (iii) allocation by physical property (e.g. fibre length). Key pitfall (VCS 2.0 literature, Sustainability 2013): the correction depends critically on the chosen point of substitution along the value chain — omitting sorting/pre-treatment steps overcorrects. For a biopolymer that loses mechanical integrity, β well below 1 (or substitution against a low-grade material such as wood flour or aggregate rather than virgin biopolymer) is appropriate, and should be sensitivity-tested.

Cross-cutting §4 — Reporting C and D: separate, combined, or range?

Report Modules C and D separately — this is mandatory under EN 15804+A2 (modules cannot be added) and is best practice. Module D is a clearly labelled, potential, beyond-boundary benefit subject to high uncertainty (future technology, markets). Recommended presentation: report C1–C4 as the committed impact; report D separately; and present the C-plus-D-range across EoL scenarios to show the decision space. Folding D into the headline number risks greenwashing.

Summary Comparison Table

Approach name	Modules affected	Treatment of biogenic carbon	Applicability to bio-based materials	Key limitations
Cut-off (100:0 / recycled-content)	C (burdens only; no D credit)	Biogenic C released at EoL (+1) in C; net zero; recyclate leaves burden-free	High — most defensible for novel materials with no market; conservative	No recycling/reuse credit even where genuine benefit exists; can understate circularity
Avoided burden / substitution (0:100)	C burdens + D credits	−1/+1 in studied system; virtual emission when C exits to other system	Moderate — needs a credible substitute; risky where market absent	Credits speculative substitutes; double-counting risk; sensitive to substitute choice
50/50 method	C + D (credits halved)	Net-zero biogenic; balanced across two cycles	Moderate — compromise when supply/demand uncertain	Arbitrary equal split; weak physical/market justification
System expansion	C + expanded boundary	Tracks full biogenic balance across expanded system	High for energy recovery and reuse	Data-heavy; choice of avoided process drives result
Circular Footprint Formula (CFF / PEF / EF 3.1)	A (R1), C and D combined via formula	Net-zero biogenic; virtual emission rule; GWP-GHG zeroes biogenic CO_2	Moderate — no default A-factor for novel biodegradables (defaults to A=0.5)	Mis-handles open-loop in EN 15804 simplification; B=0 energy assumption often wrong; parameter-heavy
Closed-loop allocation (ISO 14044)	A/C/D netted	Net-zero; carbon stays in loop	Only where properties maintained — rare for biopolymers	Real closed loops seldom exist for these materials

Approach	Modules affected	Treatment of biogenic carbon	Applicability to bio-based materials	Key limitations
Loss-of-quality / Value-Corrected Substitution	C + D (corrected credit)	Net-zero biogenic	High — explicitly captures downcycling	Sensitive to chosen point of substitution and price data

Recommendations

Stage 1 — Baseline (do now). Build Module C on an **attributional cut-off model** using the realistic current fate of mixed construction waste in your region (incineration with energy recovery, or landfill where dominant). Apply your existing Module A biogenic-carbon method identically in Module C: book the +1 release in C3 (incineration) or C4 (landfill) so biogenic carbon nets to zero. Report C1–C4 as the committed result. This is the most defensible primary number because it claims no credit from infrastructure that does not exist.

Stage 2 — Module D scenarios (parallel, clearly separated). Compute Module D credits for (a) energy recovery (avoided regional average grid electricity + heat), (b) composting (fertiliser+peat substitution at realistic ~50% rates, MFE 0.2–0.4 for N), and (c) mechanical/open-loop recycling with a value-correction factor $\beta<1$. Label all as contingent, beyond-boundary potentials. Never add D to C in the headline.

Stage 3 — Sensitivity & uncertainty (decisive). Treat as primary sensitivity parameters: the substituted energy/material in D; the quality/value-correction factor; landfill gas-capture efficiency; compost substitution rate and process CH_4/N_2O; and the biogenic-carbon convention (run −1/+1 net-zero as base, and a transparent biogenic-storage variant only as an illustrative, non-compliant sensitivity). Use scenario analysis as the deliverable, not a single score — Bisinella et al.'s "explorative approach to LCA identifies robust solutions across possible scenarios."

Benchmarks that would change the recommendation: If a dedicated take-back/recycling or industrial composting stream becomes demonstrably available for the material at commercial scale, shift the baseline from cut-off to a CFF or substitution model with the appropriate A-factor and document the market evidence. If the panel's binder/additives are substantially fossil-derived, fossil GWP in C3 (incineration) becomes material and may make recycling/landfill preferable on GWP.

Method choice verdict: For a research-context LCA of a novel bio-based panel, use **cut-off as the compliant, conservative baseline + substitution/avoided-burden in a separated Module D + mandatory scenario/sensitivity analysis.** Avoid relying on the CFF as the primary method here because no default A-factor exists for novel biodegradable materials (it would default to A=0.5, an unjustified equilibrium assumption) and its EN 15804 simplification mishandles open-loop recycling — though you may report a CFF variant for EPD comparability.

Key References

- CEN. EN 15804:2012+A2:2019 (and AC:2022 corrigendum) — Sustainability of construction works — Environmental Product Declarations — Core rules for the product category of construction products. (Module C1–C4/D structure; biogenic −1/+1 and clause 5.4.3 no-storage rule; informative CFF-derived Module D formula.)
- ISO 14040:2006 and ISO 14044:2006 (incl. Amendment 2:2020) — LCA principles, framework, requirements and guidelines (allocation hierarchy; clause 4.3.4.3 recycling).
- European Commission. Recommendation (EU) 2021/2279, PEF method, Annex I §4.4.8 and Annex C (Circular Footprint Formula; default A and R values). JRC: Zampori & Pant (2019), *Suggestions for updating the Product Environmental Footprint (PEF) method*, JRC technical report.
- Allacker, Mathieux, Pennington & Pant (2017). The search for an appropriate end-of-life formula for the EC Environmental Footprint initiative. *Int. J. Life Cycle Assess.* doi:10.1007/s11367-016-1244-0.
- Laurent et al. (2014). Review of LCA studies of solid waste management systems — Part I & II. *Waste Management* 34(3):573–588, 589–606.
- Bisinella, Schmidt, Varling, Laner & Christensen (2023/2024). Waste LCA and the future. *Waste Management*.
- Sandin & Peters (2018). Environmental impact of textile reuse and recycling — A review. *J. Cleaner Production* 184:353–365.
- Boldrin et al. (2010). Environmental inventory modelling of the use of compost and peat in growth media preparation. *Resources, Conservation and Recycling*, and Andersen et al. (2010), Substitution of peat, fertiliser and manure by compost in hobby gardening. *Waste Management*.
- Nicholson et al. (2009). End-of-life LCA allocation methods: open-loop recycling impacts on robustness of material selection decisions. MIT.
- Ekvall (1994/2000) and Klöpffer (1996) — open-loop allocation criteria and 50/50 method.
- Anshassi, Sackles & Townsend (2021). Review of LCA assumptions impacting whether landfilling or incineration results in less GHG. *Resources, Conservation and Recycling*.
- Astrup, Tonini, Turconi & Boldrin (2014). LCA of thermal waste-to-energy technologies: review and recommendations. *Waste Management* 37:104–115.
- Galluccio, Tamke, Nicholas, Svilans, Gaudillière-Jami & Ramsgaard Thomsen (2025). Material Stories: Assessing Sustainability of Digital Fabrication with Bio-Based Materials Through LCA. doi:10.1007/978-3-031-69626-8_3 (open access; CITA Radicant panels).
- Heijungs et al. (2021). System Expansion and Substitution in LCA: A Lost Opportunity of ISO 14044 Amendment 2. *Frontiers in Sustainability*.
- Hottle et al. / Bioplastics-at-EoL review (2022). A review of bioplastics at end-of-life. *Resources, Conservation and Recycling*, and the building biogenic-carbon overview, *Buildings & Cities* (doi:10.5334/bc.46).

Caveats

- EN 15804+A2 normative clause text (5.4.3 on storage; C.2.4 on the +1 factor) is paywalled; clause numbers and wording here come from authoritative secondary syntheses and PCRs citing those clauses — verify against the purchased standard before publication.
- The AC:2022 corrigendum is understood to make editorial/technical corrections without altering the −1/+1 net-zero biogenic principle; confirm directly if a precise corrigendum citation is needed.
- The A=0.5 default for unlisted novel/biodegradable materials is an inference from the PEF decision hierarchy, corroborated by a documented Annex C gap (Int. J. LCA carpet-recycling case study), not an explicit "biodegradable" clause.
- Active methodological debates flagged: (i) landfill vs incineration GWP ranking depends on gas-capture assumptions; (ii) whether biogenic temporary/permanent storage should ever be credited (EN 15804 says no; dynamic-LCA literature argues timing matters); (iii) whether substitution is legitimate in attributional LCA (ISO 14044 Amendment 2 left this unresolved); (iv) appropriate marginal vs average energy substitute.
- Quantitative datapoints for specific biopolymers (PLA) are used illustratively; primary EoL inventory data for seagrass/bark/pea-protein 3D-printed panels are essentially absent in the literature, which is itself a key finding — expect high uncertainty and rely on sensitivity analysis.